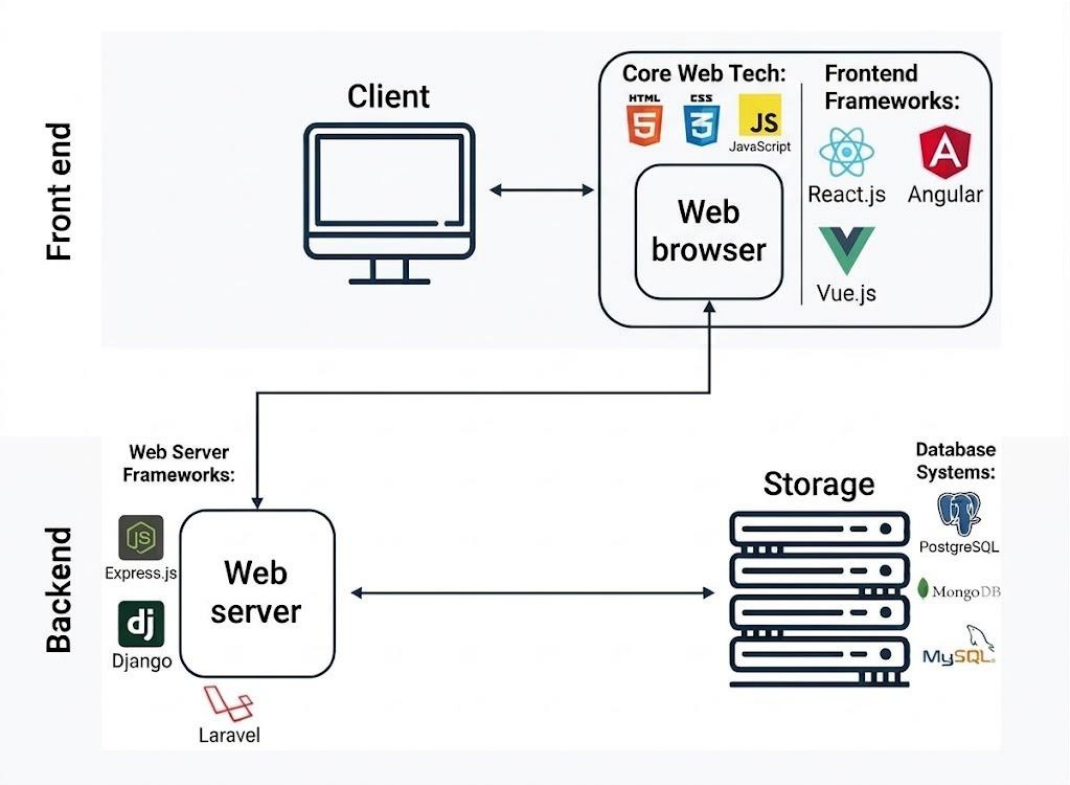
Web Development

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Web Development (Recap)



Course Roadmap





Human Computer Interaction

Human: the person that's using the system.

Computer: the **machine** or the network of machines that **run** the system.

Interaction: the **interface** that **represents** the **system** for the **user**.



Human Computer Interaction

Human Computer Interaction is the **study**, **planning**, and **design** of how **people** and **computers work together** so that a person's **needs** are satisfied in the **most effective way**.

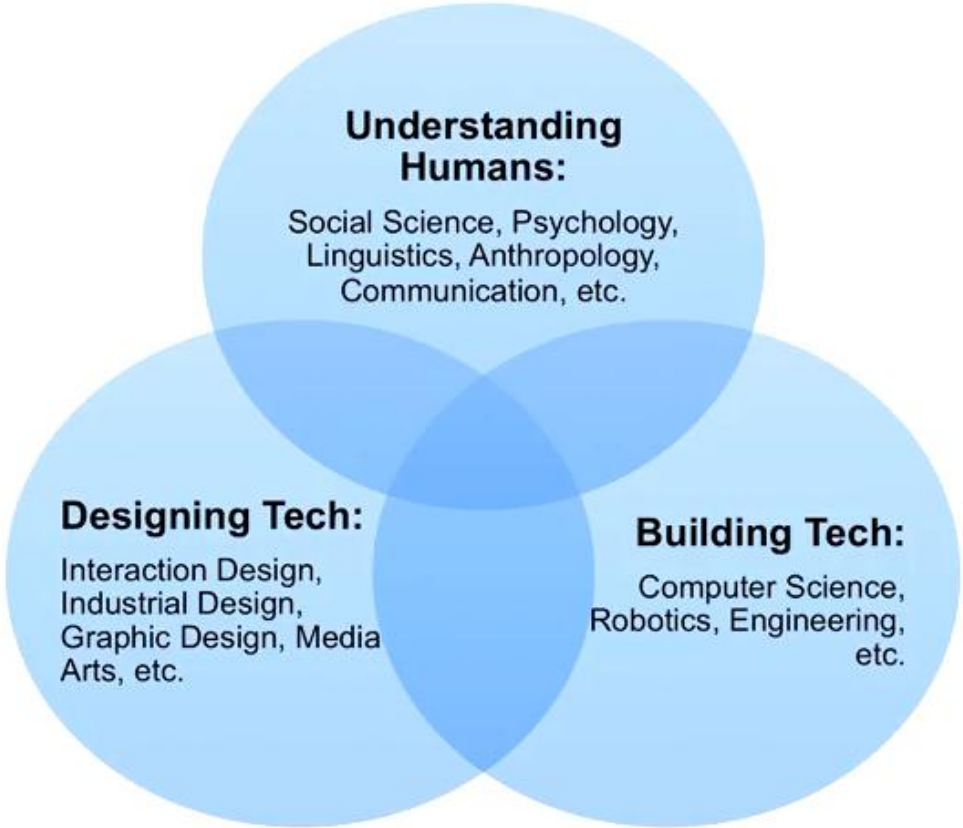


Human Computer Interaction

- HCI designers must consider a variety of factors:
  - What people **want** and **expect**
  - What physical **limitations** and **abilities** people possess
  - How their **perceptual** and information processing systems work
  - What people find **enjoyable** and **attractive**

- Technical **characteristics** and **limitations** of the computer **hardware** and **software** must also be considered.

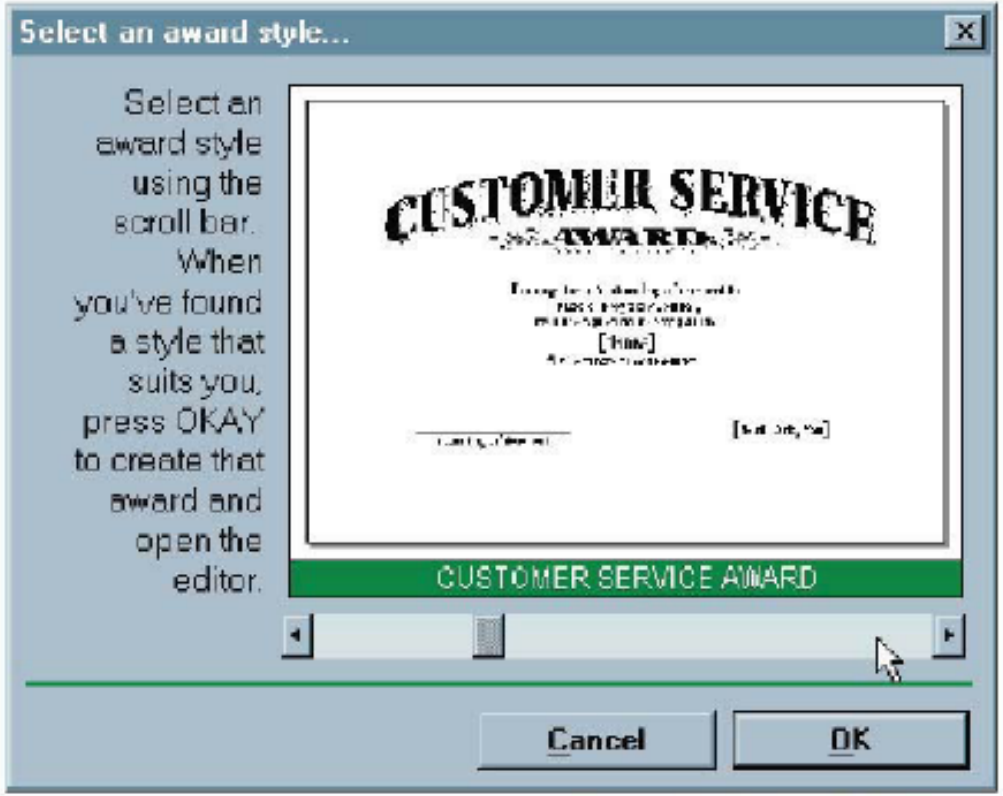
Human Computer Interaction Relevant Disciplines



Applications of HCI

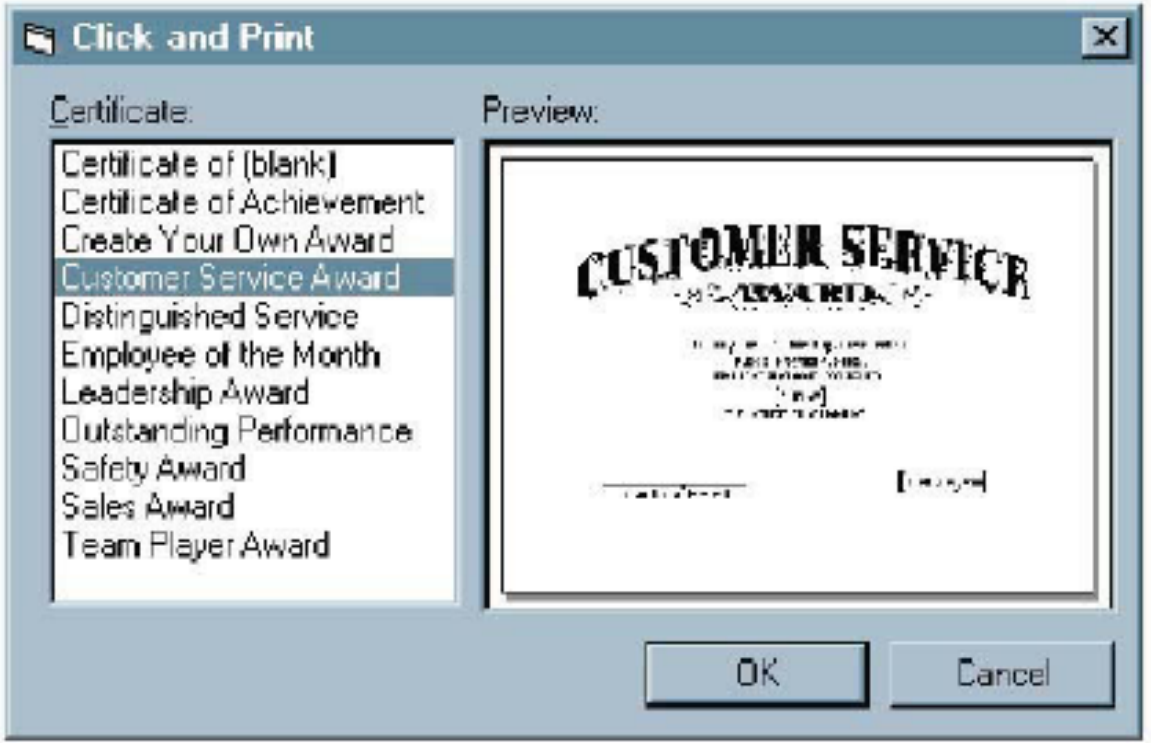
- Everyday Life: ATM machines, information Kiosks, railway ticket selling machines, and vending machines, ... etc
- Education
- Industry and business
- Virtual Reality
- Voice search, such as Amazon Alexa and Google Voice Search
- Cybersecurity & Usable Privacy: Secure authentication interfaces.

UI Hall of Shame



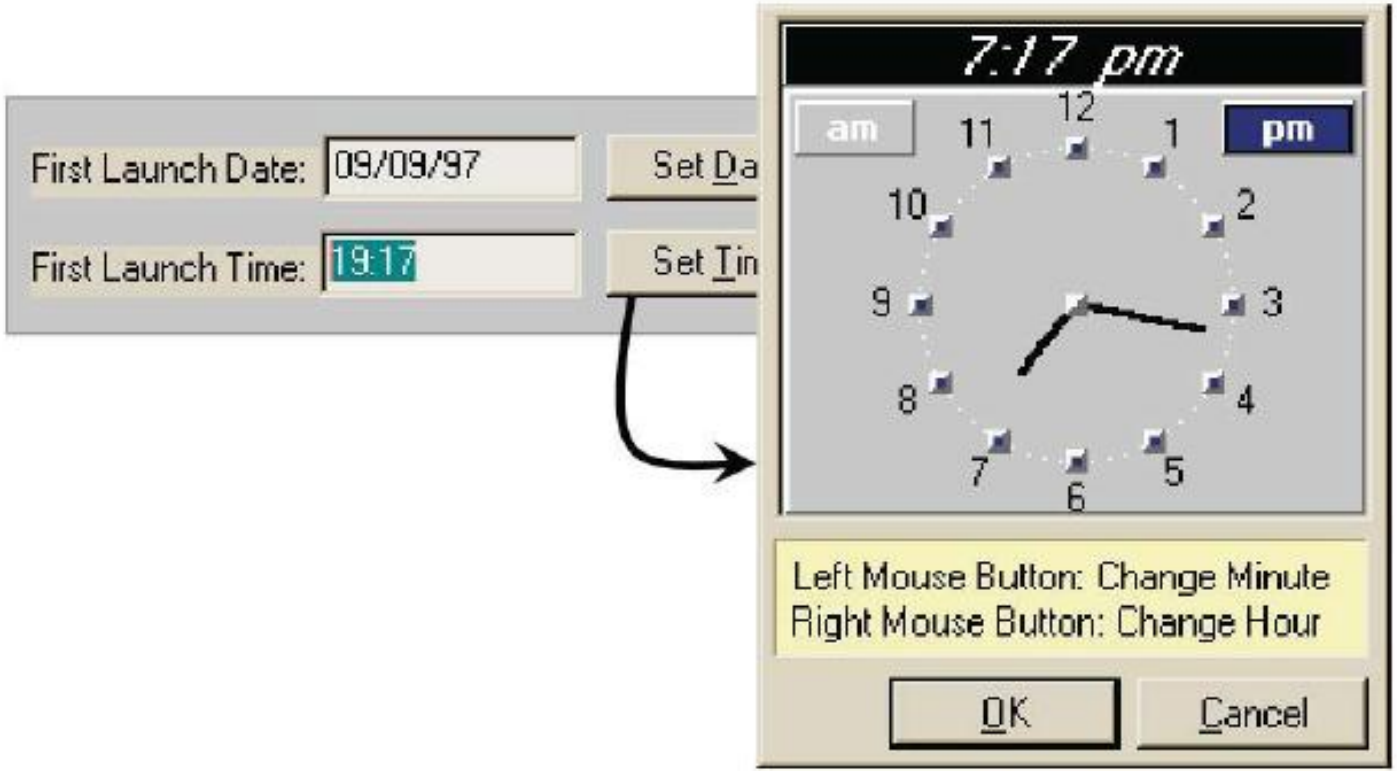
Source: Interface Hall of Shame

Redesigned



Source: Interface Hall of Shame

More UI Hall of Shame

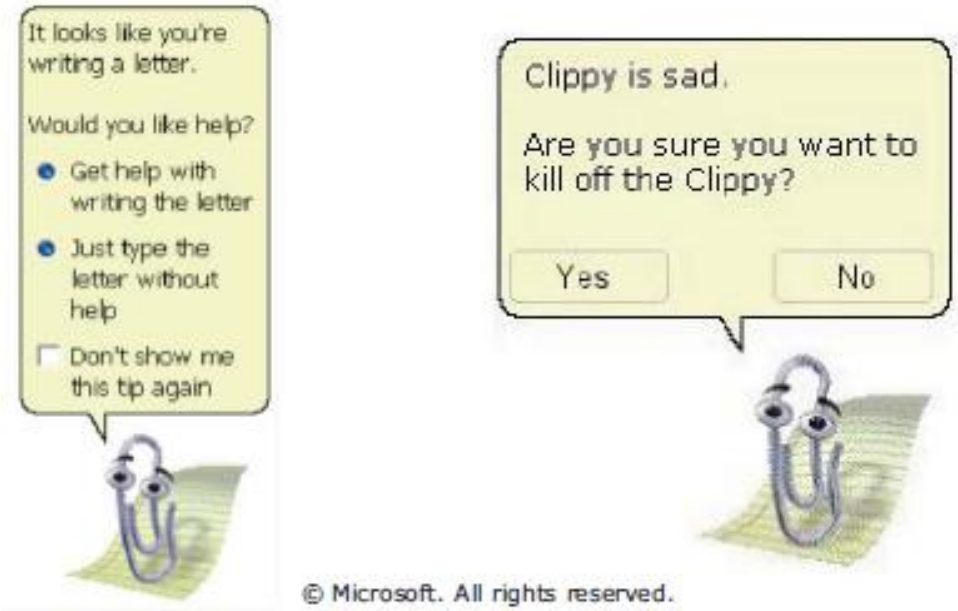


Source: Interface Hall of Shame

UI Hall of Shame



UI Hall of Shame/Fame



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Good Design

- Proper interface design will provide a **mix of well-designed input and output mechanisms** that satisfy the user's **needs**, **capabilities**, and **limitations** in the most effective way possible.

- The best interface is one that it **not noticed**, one that permits the user to **focus** on the **information** and **task** at hand, **not** the **mechanisms** used to present the information and perform the task.

Importance of User Interface

- **Usability** strongly affects how software is **perceived**, because the user interface is the means by which the software presents itself to the world.

- A Harris poll (reported in the Wall Street Journal 11/8/05) found that **ease of use** (61%), customer service (58%), and no hassle installation (57%) are the most important factors US adults consider when purchasing a new technology product.

Usability

Usability measures **how well** users can **use** the system's functionality.

1

Learnability

Is it easy to **learn**?

2

Efficiency

Once learned, is it easy to use?

3

Safety

Are errors **few** and **recoverable**?

Learnability

Learnability: Easy to Learn

learnability and **memorability** – making interfaces **easier** for **new users** to **learn**, and for **casual users** to **remember**.

Topics

- Interaction Styles
- Conceptual Models of UI
- Learning Principles

Recognition vs. Recall

- **Recognition:** remembering with the help of a visible ***cue***
  Aka: Knowledge in the world

- **Recall:** remembering with ***no help***.
  Aka: Knowledge in the head

- Recognition is much easier

Interaction Styles

- Command Language
- Menus & forms
- Direct Manipulation

Command Language

- The earliest computer interfaces
- Users ***type in commands*** in an artificial language

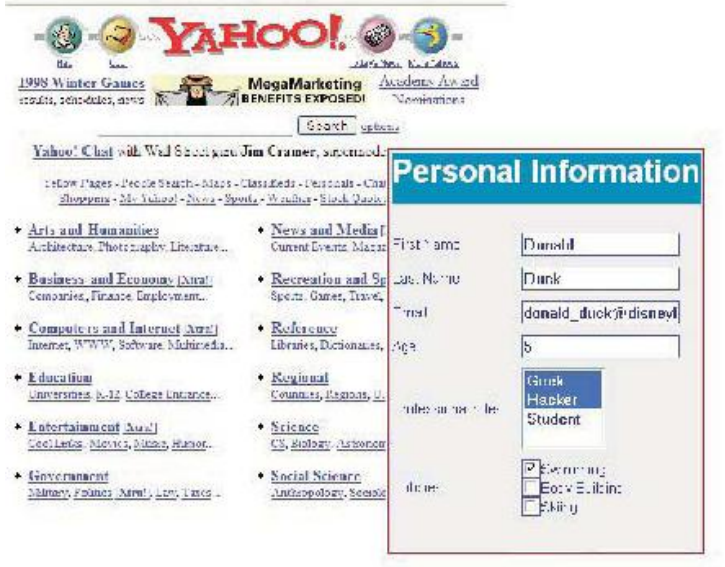
```
ls -l *.java
```
Unix shell

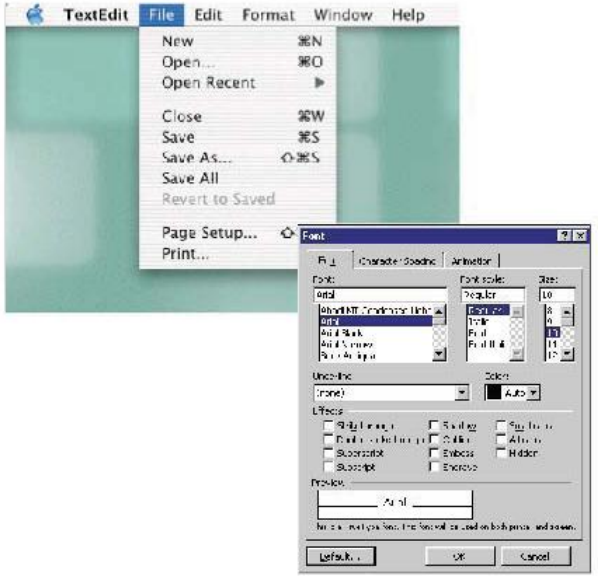
```
+6.831 site:mit.edu
```
search engine query

```
http://www.mit.edu/admissions/
```
URL

Menus and Forms

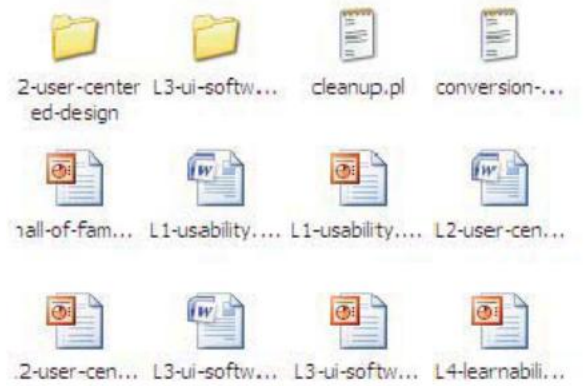
- User is **prompted** to choose from **menus** and **fill in forms**.
- Most graphical user interfaces have some kind of menu/forms interaction, such as a menubar and dialog boxes.



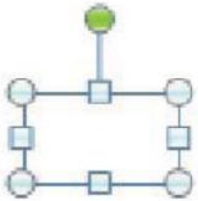


Direct Manipulation

- User ***interacts*** with **visual representation** of data objects.



Files & folders on desktop



Selection handles



Scrollbar



Dropping Pins

Direct Manipulation

1- **Continuous visual representation:** The representation may be **verbal (words)** or **iconic (pictures),** but it's **continuously displayed**, **not** displayed on **demand**.

Ex: Icons representing files and folders on desktop
Graphical objects in a drawing editor
Email messages in your inbox.

2- The user interacts with the visual representation using **physical actions** or **labeled button presses**. (clicking and dragging)

Direct Manipulation

3- The effects of actions should be

- **Rapid** visible as quickly as possible
- **Incremental** you can drag the scrollbar thumb a little or a lot, and you see each incremental change
- **Reversible** you can undo your operation
- **Immediately visible** the user doesn't have to do anything to see the effects.

Why is direct manipulation so powerful?

- It **exploits perceptual** and **motor skills** of the human machine

- Depends **less on linguistic skills** than command or menu/form interfaces.

Comparison of interaction styles

	Command Line	Menus/Forms	Direct Manipulation
1- Knowledge in the head vs Knowledge in the world	***Requires significant learning***. Users must put a lot of knowledge into their heads in order to use the language, by reading, training, practice, etc.	***Puts much more information*** into the world, i.e. into the interface itself.	Also has ***information in the world***, delivered by the affordances, feedback, and constraints of the visual object.
	Since recognition is so much easier than recall, this means that **MF and DM is much more learnable and memorable than CL.**		
2- User Type	Generally **better** for ***expert users***, who keep their knowledge **active and who are willing to invest in training and learning in exchange for greater efficiency.**	**MF and DM are generally better for beginners and infrequent users.**	
3- Accessibility	CL and MF interfaces are **more textual**, so they are ***easier for vision-impaired*** users to read with screen readers.		**DM interfaces are *much harder* for these users**

Topics

- Interaction Styles
- Conceptual Models of UI
- Learning Principles

Model

- A **model** of a system is a way of describing **how the system works**.
- It specifies
    - What the **parts** of the system are.
    - **How** those **parts interact** to make the system **do** what it's **supposed to do**.

- Learning a new system **requires the user to build a mental model of how the system works.**

- **Learnability** can be strongly **affected** by **difficulties** in **building that model**.

Models in UI Design

There are actually **several models** you have to worry about in UI design:
1- The **system model** (implementation model) is **how** the **system actually works**.

2- The **interface model** (or manifest model) is the model that the **system presents** to the user through its **user interface**.

3- The **user model** (or conceptual model) is **how the user thinks the system works**.

- The **interface model** might be quite **different** from the **system model**. **Ex:** Cell Phone

- Finally, the **user model** isn't **software** at all; it's all in the **user's mind**.

Models in UI Design (Cont.)

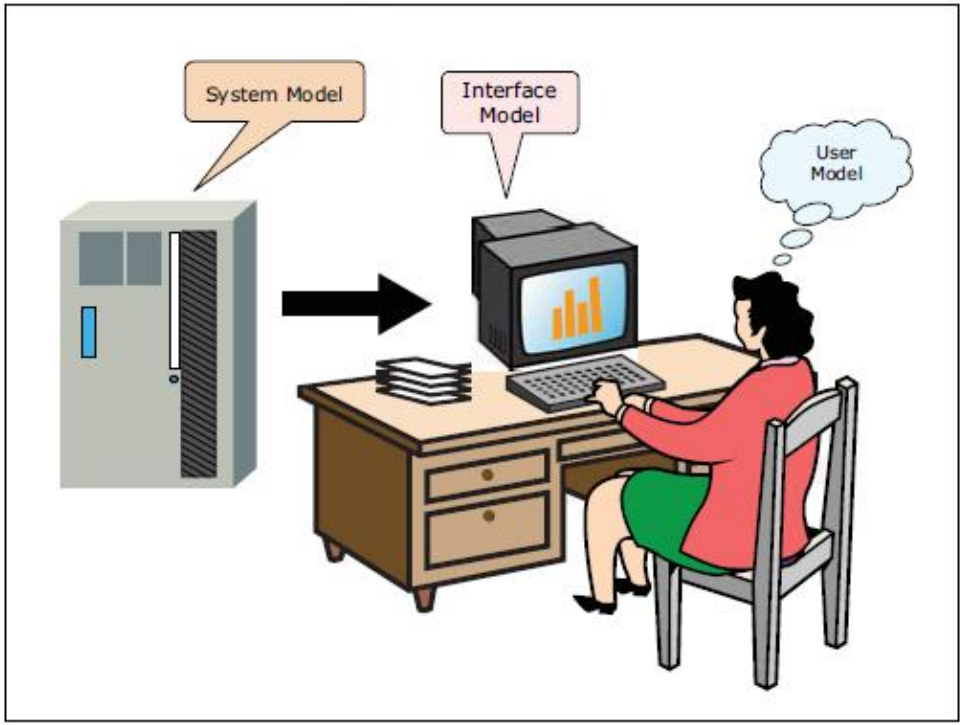


Image by MIT OpenCourseWare.

Models in UI Design (Cont.)

- As a software engineer, you should be quite familiar with this notion:
    - A ***module interface*** offers a certain model of operation to clients of the module, but its ***implementation*** may be ***significantly different.***

- In user interface design, we value it primarily for other reasons: the ***interface model*** should be ***simpler*** and more closely ***reflect*** the ***user's model of the actual task.***

Interface Model Hide System Model

- The Interface model should be:
  - Simple
  - Appropriate (Reflects user model of the task)
  - Well Communicated

Topics

- Interaction Styles
- Conceptual Models of UI
- Learning Principles

Learning Principles

- Consistency
- Affordance
- Feedback
- Information Scent

Metaphor

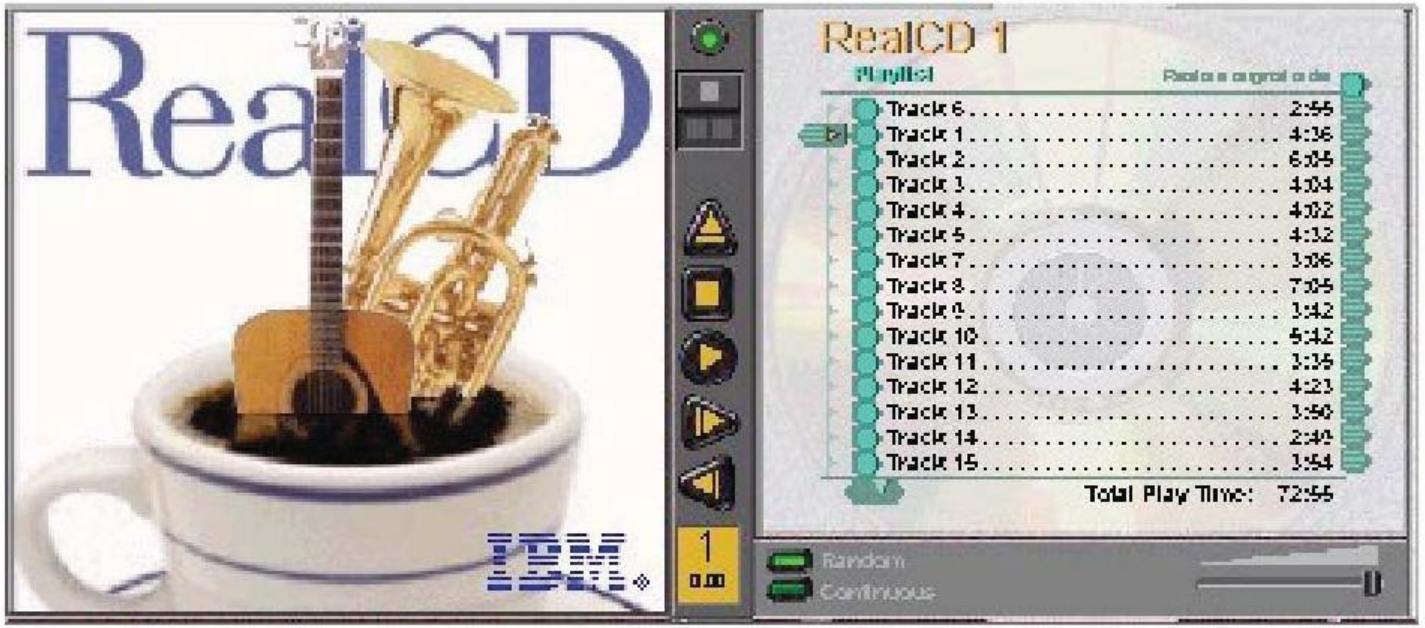
Metaphor: an analogue in the real world.

Metaphors are one way to make an interface more learnable, since **users** can **make guesses** about **how it will work based on what they already know about the interface's metaphor.**
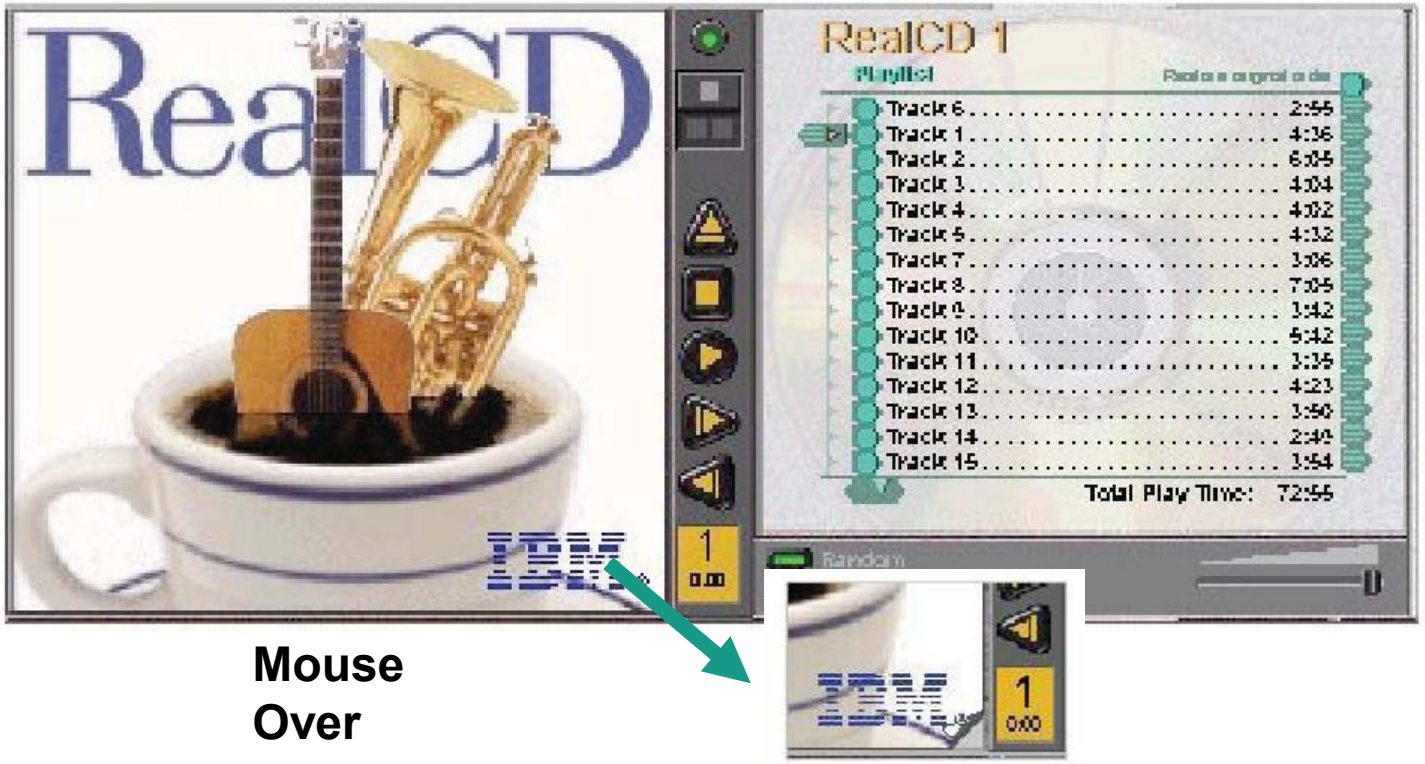
UI Hall of Shame/Fame?



UI Hall of Shame/Fame?



UI Hall of Shame/Fame?



Lessons Learned

This example has several lessons:
1- That interface metaphors can be horribly **misused.**

2- That the presence of a metaphor does not at all guarantee an **easy-to-learn** user interface.

3- Unrelated to metaphor – ***that beautiful graphic design doesn't equal usability***, and that graphic designers can be just as blind to usability problems as programmers can.

- **Metaphor** is not the only way to achieve learnability. In fact, it's probably the hardest way.

Metaphors

Advantages

- Highly **learnable** when **suitable**
- **Hooks** into user's **existing mental models** very easily

Dangers

- Often **hard** for designers to **find** (use one if you have)
- May be **deceptive** (leading users to infer behavior that interface doesn't provide)
- May be **constraining** (desktop folders)
- Metaphor is usually **broken** somewhere (typewriter metaphor)



Desktop metaphor



Trashcan metaphor



Typewriter metaphor

Learning Principles

- Consistency
- Affordance
- Feedback
- Information Scent

Consistency

- The Principle of “Least Surprise”, which basically means that you ***shouldn’t surprise*** the user with the ***way a command*** or ***interface object works***.

- Similar things should look, and act, in similar ways.
- Conversely, different things should be visibly different**.**

Kinds of Consistency

There are three kinds of consistency you need to worry about:
1- Internal consistency within your application
2- External consistency with other applications on the same platform;
3- Metaphorical consistency with your interface metaphor or similar real-world objects.

The RealCD interface has problems with both metaphorical consistency (CD jewel cases don't play; you don't open them by pressing a button on the spine; and they don't open as shown), and with external consistency (the player controls aren't arranged horizontally as they're usually seen; and the track list doesn't use the same scrollbar that other applications do).

Kinds of Consistency







Internal: with itself

External: with other UIs or the real world

Metaphorical: with the chosen interface metaphor

Natural Mapping: Consistency of Layout

- When possible, the **physical arrangement** of **controls** should **match arrangements** of **function**.

- Best mapping is **direct**, but **natural mappings don't have to be direct** if they have an **easy mental model.**

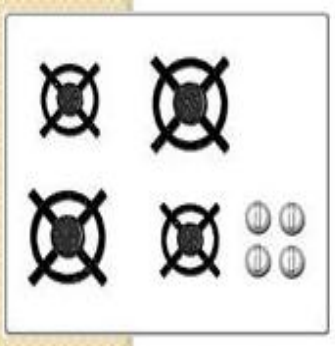
- Consider the spatial arrangement of a light switch panel. How does each switch correspond to the light it controls? If the switches are arranged in the same fashion as the lights themselves. It is much easier to learn which controls which light.

- Direct mappings are not always easy to achieve, since a control may be oriented differently from the function it controls. Light switches are mounted vertically, on a wall; the lights themselves are mounted horizontally, on a ceiling. So the switch arrangement may not correspond *directly* to a light arrangement.

Natural Mapping: Consistency of Layout

Other good examples of mapping include:
Stove burners. Many stoves have four burners arranged in a square, and four control knobs arranged in a row. Which knobs control which burners?

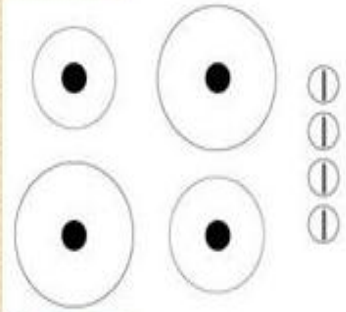
Car turn signals. The turn signal switch in most cars is a stalk that moves up and down, but the function it controls is a signal for left or right turn. So the mapping is not direct, but it is yet natural.



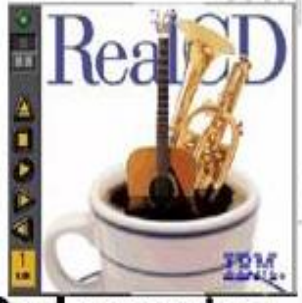
Direct mapping



Natural mapping



Bad mapping



Bad mapping



Direct mapping

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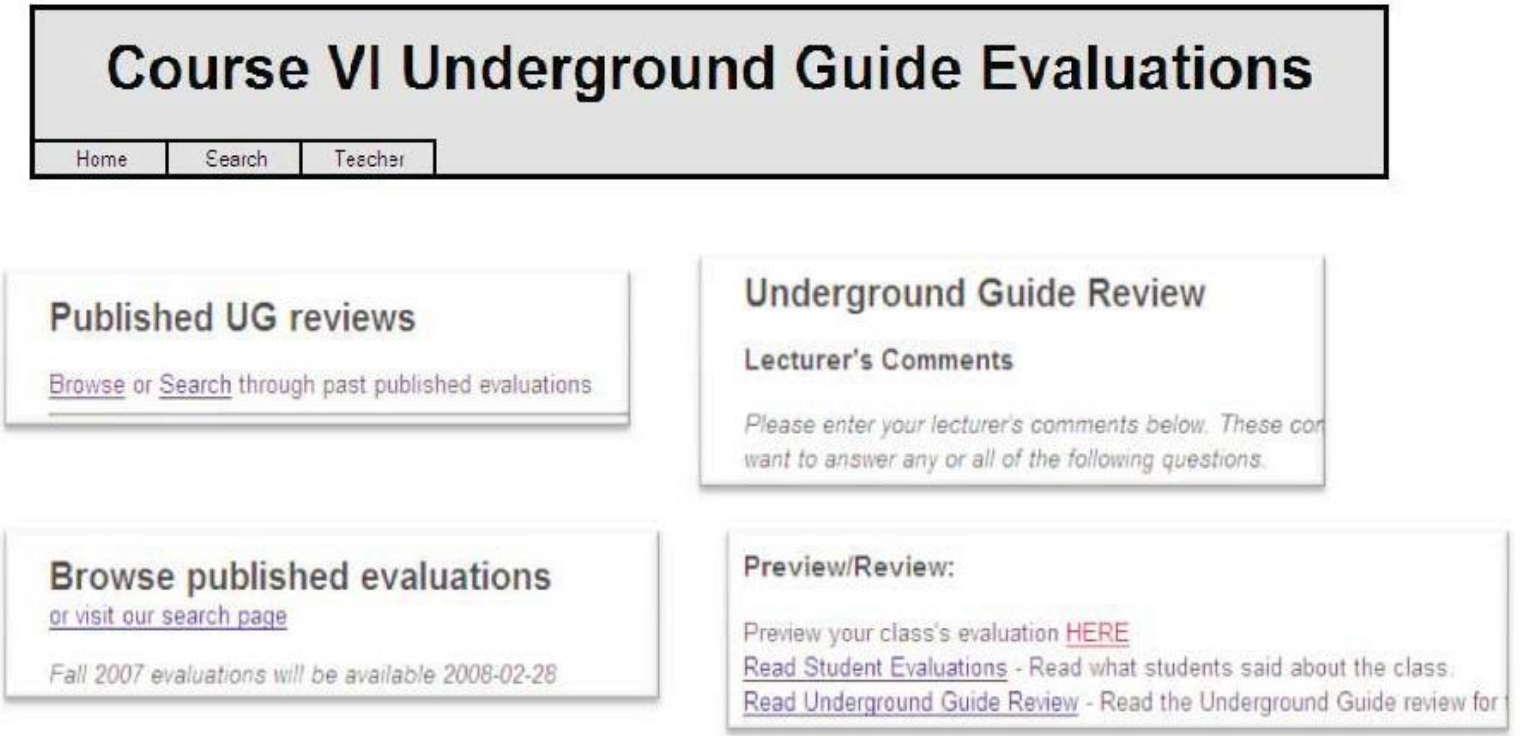
Internal Consistency in Wording

- Use the **same terms throughout** your user interface.

Ex: If your interface says •share price in one place, •stock price in another, and •stock quote in a third, users will wonder whether these are three different things you're talking about.

- **Don't get creative** when you're writing text for a user interface; keep it **simple** and uniform.

Internal Consistency in Wording



External Consistency of wording

- Speak the user's language.
- Technical terms should be avoided.
- Use of technical term reflects aspects of the **system model creeping up** into the **interface model**, unnecessarily.



- Use domain-specific terms when it is specific to the application domain and the expected users are domain experts.

Learning Principles

- Consistency
- Affordance
- Feedback
- Information Scent

Affordance

Affordances are **how** an **interface** communicates **nonverbally**, telling you how to **operate** it.



Actual affordance

The door can be pushed.

Perceived affordance

The handle suggests pulling.

Mismatch = predictable mistake

Affordances

- **Perceived** and **actual properties** of a thing that **determine** how the **thing** could be **used**.









Dr Huda Amin
Information Systems Departmen

- Perceived vs. Actual



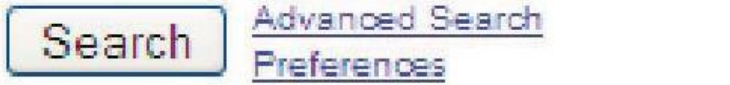


Affordances

- Affordances are **how** an **interface** communicates **nonverbally**, telling you how to **operate** it.
- The parts of a user interface should **agree** in **perceived** and **actual** affordances.
- Affordances are **learned** from **experience**.

Use Appropriate Affordances

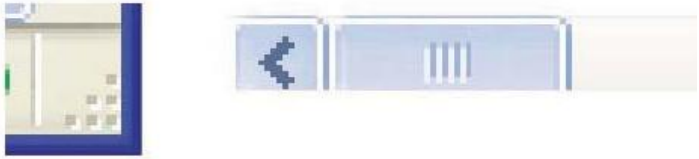
- Buttons & links



- Drop-down arrows



- Texture



- Mouse cursor



- Highlight on mouseover



Learning Principles

- Consistency
- Affordance
- Feedback
- Information Scent

Feedback

- How the system **changes visibly** when you **perform** an **action**.
- When the user successfully makes a part work, it should appear to respond.

  Push buttons depress and release.

  Scrollbar thumbs move.

  Dragged objects follow the cursor.

Feedback doesn't always have to be visual. **Audio** feedback – like the clicks that a keyboard makes – is another form.

Actions should have Immediately Visible Effects

- Low-level feedback

  - Shows that the interface at least ***took notice*** of the user's input, and is responding to it



- High-level feedback

  - Model State Changes

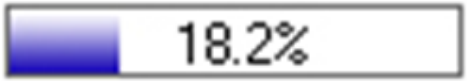
  - New web page starts loading

Response Time

< 0.1 s: seems instantaneous

0.1-1 s: user notices the delay

1-5 s: display busy indicator  

> 1-5 s: display progress bar 

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Feedback Visibility Depends on Locus of Attention

- ***How to make the set of actions visible***.

- Spotlight of attention: attention focuses on one input channel (e.g. area of visual field) at a time.

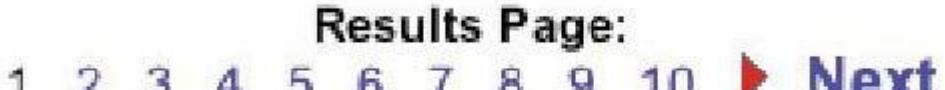
- Does the user's locus of attention include:
  - Caps Lock light on keyboard
  - Status bar
  - Menu bar?
  - Mouse Cursor

Visible Navigation State

- Breadcrumbs



- Pagination



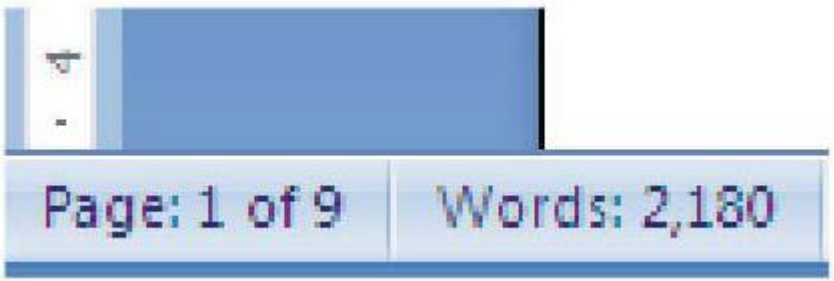
- Tabs



Visible Model State

- Continuous visual representation of model

- What to **visualize** should be **guided** by the **user's tasks**

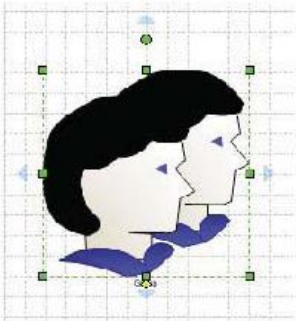


Visible View State

- "view state" is the **current state** of the **user's interaction** with the interface.
  - Selection highlight



  - Selection handles



  - Drag & drop



Useless Feedback vs. Useful Feedback

- Feedback is important, but don’t overdo it.



Learning Principles

- Consistency
- Affordance
- Feedback
- Information Scent

Information Scent

- **Information foraging theory**

  - Humans gathering information can be modeled like animals gathering food

  - Constantly evaluating and making decisions to maximize information collected against cost of obtaining it.

- Information scent

Cues on a **link** that **indicate** how **profitable** it will be to **follow** the **link** to its destination.

Give Good Information Scent

- A **Link** should **smell** like the **content** it **leads to**.

- Ex: Date, Time, Language and regional options.
  Printers and Other Hardware in control panel.

- **Quality** of information scent **depends** on the **user's particular goal.**

Ex: Shopping site for Music and Movies. (Movie sound track?)
Solution: put it in both categories or provide "see also"

Add dates for prices

Select dates

Minimum stay: 30 nights

CHECK-IN
26/04/2025

CHECKOUT
Add date

April 2025

Mon	Tue	Wed	Thu	Fri	Sat	Sun
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

30-night minimum

May 2025

Mon	Tue	Wed	Thu	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Clear dates

Close

macula
gether

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Topics Covered

- Interaction Styles
- Conceptual Models of UI
- Learning Principles
  - Consistency
  - Affordance
  - Feedback
  - Information Scent

Sources

- Human Computer Interaction, Stanford University
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- Don Norman, the Design of Everyday Things,2013
- The Essential Guide to User Interface Design 2nd Edition, Wilbert O. Galitz
- HCI Classes

Thank You